

money

- ❑ **rai stones** in Yap Island
- ❑ **cowry shells** in Kenya
- ❑ **barley** in Mesopotamia
- ❑ **paper money** in China, backed by metal
- ❑ **fiat** currency
- ❑ **digital** money
- ❑ **cryptocurrency**



gold

- Store of value for **8000** years
- Corrosion resistant
- Fungible
- Rare/Scarce (165,000 metric tons)
- Pretty
- USA left gold standard in 1933
- (USA banned personal ownership of gold in 1933 via Executive Order 6102)
- 8 Trillion** USD market cap



Precursor Digital Currencies

1

David Chaum's **Digital Cash**

- Paper published in 1983, used PKI to secure transactions
- Mark Twain Bank in Saint Louis MO 1995 – 1998
- Never widely adopted on internet, overtaken by VISA
- Bankrupt in 1998

2

e-gold

- Backed by gold
- Mobile transactions (on Palm Pilot)
- Popular with hackers and criminals
- In 2008, 3 directors pled guilty to running an illegal money transmitter

What makes bitcoin different to these other digital currencies?



Bitcoin – a decentralised cryptocurrency

- ▮ Decentralised
- ▮ No controlling entity – no offices to raid, no creator to arrest
- ▮ No governance
- ▮ Who created it?

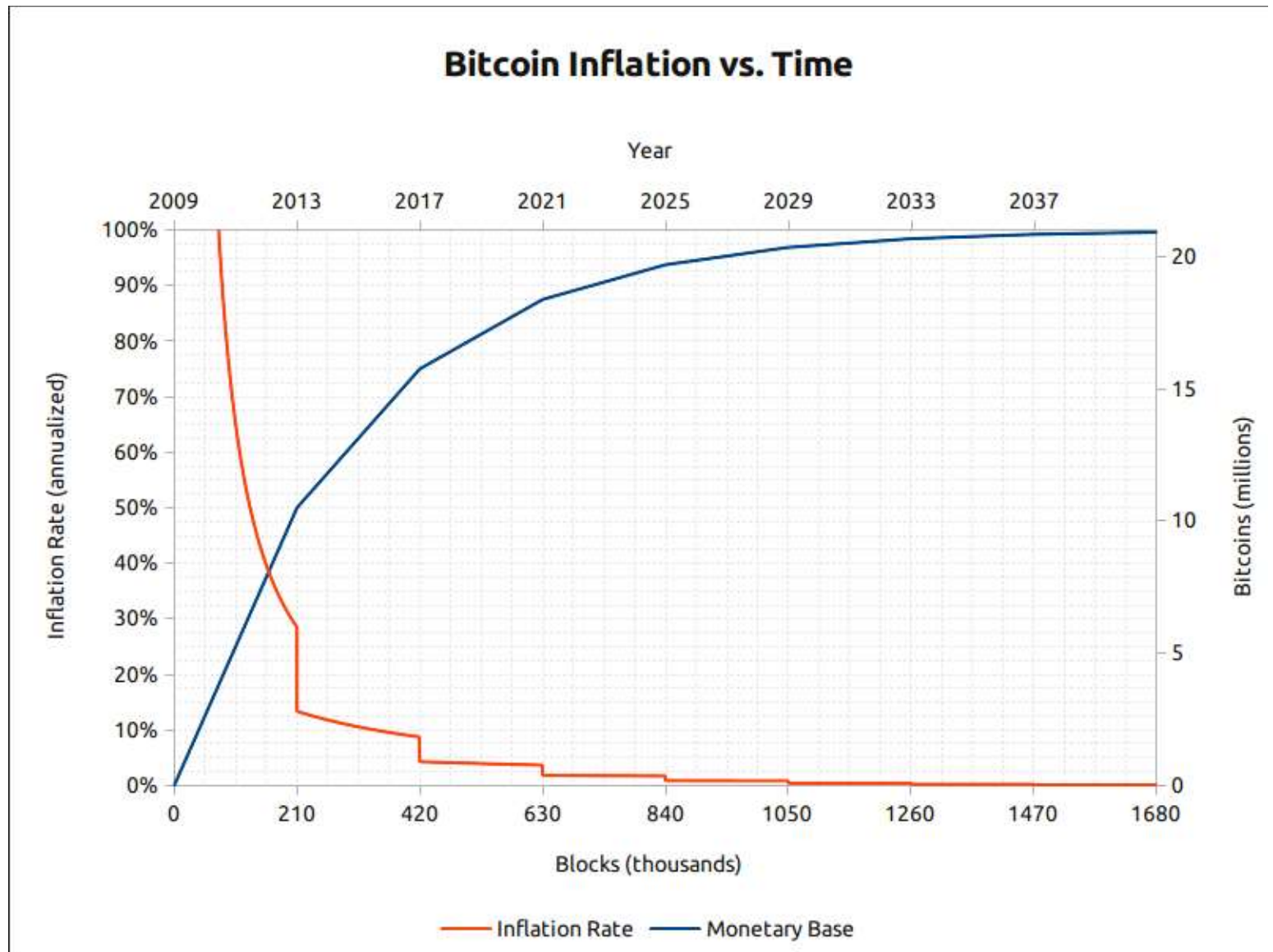
Satoshi Nakamoto

- ❑ Wrote the white paper
- ❑ Wrote the first implementation – still under development today
- ❑ Not his/her real name
- ❑ All interactions conducted over Tor to protect IP address
- ❑ Stepped away from ecosystem after two years
- ❑ Has not spent any of the 1,000,000 coins mined (current value: 11 Billion USD)



Bitcoin Issuance over time

- Bitcoin is issued by mining a block
- Block reward is 6.25 BTC + fees
- There will only ever be 21 million BTC issued
- The genesis block issued 50 BTC
- This halves every 4 years, currently only 6.25 BTC are issued in each block
- All BTC will be issued by ~2140





China bans bitcoin

- ❑ China bans ICOs
 - ❑ China bans exchanges
 - ❑ China bans OTC exchanges
 - ❑ China bans mining?
 - ❑ China bans bitcoin?
-
- ❑ What have Gmail, Youtube, Twitter, Facebook, Github, Instagram, Slack, Vimeo, DuckDuckGo, Tumblr, Reuters, Flickr and the New York Times got in common?
 - ❑ What happens to bitcoin if it is banned in China?



Properties of Money

- Medium of Exchange
- Unit of Account
- Store of Value

Medium of Exchange

1

Fiat Examples

- Cash transaction
- Bank transfer
- Credit/Debit card interaction

2

Question: How does bitcoin compare from a privacy, cost and scale perspective?



In 2010, Laszlo bought 2 pizzas for 10,000 BTC, then worth 41 USD.

Unit of Account

- Have a standard numerical unit of measurement
- Like the dollar, yen, euro etc.
- One bitcoin is made up of 100,000,000 satoshi
- Fun fact: thanks to hyperinflation, by 2017, 1 satoshi had the same value as 1 Venezuelan Bolivar
 - <https://cointelegraph.com/news/in-bitcoin-testbed-venezuela-1-bolivar-now-equals-1-satoshi>



Store of Value

Store of Value

Question: How does bitcoin compare from a cost of storage and a maintenance of stored value compared to the above?

Question: What other items can offer store of value?

Bonds

Pension

Gold



Money must be

- ▮ Fungible (all coins are equal if they have the equal value)
- ▮ Durable (capable of being used repeatedly)
- ▮ Divisible (can be divided into smaller units)
- ▮ Portable (can be carried and transferred)
- ▮ Acceptable (others should accept your money for transactions)
- ▮ Limited in Supply (infinite supply of money = hyperinflation)
- ▮ Uniform (all coins of same value the same – e.g. not diamonds)



Does cryptocurrency fulfil these criteria?

- ❑ Do cryptocurrencies like BTC/ETH fulfil the criteria for money?
- ❑ Are there different requirements for a cryptocurrency that purely wants to be a store of value – a digital gold, rather than a coin for buying coffee?
- ❑ How important is privacy in our transactions?
- ❑ What about stablecoins like DAI/SAI, USDT, USDC?
- ❑ What about Libra?



Scale

- ▮ Assuming cryptocurrencies solve fungibility and privacy, would they be able to replace money?
- ▮ BTC is limited to 3 – 5 transactions per second, ETH 10 – 15 per second
- ▮ If Venezuela only had bitcoin, each citizen would be able to make one transaction every two weeks, not good enough for money
- ▮ Currently bitcoin blockchain is < 300GB with 1MB blocks created every 10 minutes (note: blocks can be larger than 1MB thanks to segwit – segregated witness means not counting transaction signatures as part of block size. Segregated Witness is scaling mechanism)



Platform	Transactions Per Second
Bitcoin	3 - 5
Ethereum (public)	10 - 15
Ethereum (private)	100 - 400
SWIFT	300
Paypal	500
Hyperledger Fabric (private)	700
VISA (standard)	2,000
VISA (peak)	65,000
Alipay (singles day)	255,000



Scaling Bitcoin

- ▮ Ongoing challenge
- ▮ Conference held annually to showcase new scaling mechanisms (Scaling Bitcoin, Nov 2017 Stanford University)
- ▮ Target is 1 million transactions per second
- ▮ Cannot be all scaled on chain – would require petabyte blocks
- ▮ Off-chain scaling mechanisms currently being developed
 - ▮ Lightning Network (Bitcoin, Litecoin)
 - ▮ Raiden Network (Ethereum)
- ▮ Off-chain scaling networks are 12 – 24 months away * (more later)

* *optimistic*



100 flavours of bitcoin

- ▮ 2017 brought us forks: bitcoin cash and bitcoin gold
- ▮ Ethereum forked after the DAO hack into ETH and ETC (Ethereum Classic)
- ▮ As exchanges have fiduciary duty to provide customers with their funds, these hard forks are likely to continue
- ▮ Question:
 - ▮ When a chain forks to create a new coin, does this benefit the ecosystem?



Other Cryptocurrencies

- ▮ There are close to 1000 other cryptocurrencies, not counting ICOs created on existing platforms like Ethereum
- ▮ 600 are forks of Bitcoin's code, such as Litecoin
- ▮ Some are new and don't have a blockchain, like IOTA
- ▮ Some have a particular property, like Monero's privacy
- ▮ Some are centralised, like Ripple (or IOTA)
- ▮ Some are closed source, some are open



Other
Platforms

Ethereum

Litecoin

Monero

Tezos

IOTA



Ethereum

Mines blocks every 15 seconds

Has a native currency, Ether

5 Ether released in each new block

No limit on Ether released (will require hard fork to change)

Has forked multiple times, once creating a split currency (Ethereum Classic)

Has Turing-complete smart contracts



Litecoin

Fork of bitcoin code

Mines every 2.5 minutes

Fixed issuance of 84 million litecoin

Functionally identical to bitcoin – also has segwit enabled

Cross chain atomic swaps possible with bitcoin (more later)

Silver to bitcoin's gold

Question: Is there a need for litecoin?



Monero

Has confidential transactions (hides amount, sender and receiver)

No issuance cap, but will reach limit of 1% inflation round 2022

Has 2 hard forks every year to introduce new features

Monero miners beginning to be used to replace ad revenue – potentially interesting space if it legitimises



IOTA

More of a DAG (Directed Acyclic Graph) than a blockchain

Designed for IOT use cases

Created their own hashing function (turned out to be insecure)

Application written in ternary rather than binary

10kb transaction sizes

Revealed to be highly centralised when closed course ITOA-controlled Coordinator service outage stopped transactions from being verified for 50+ hours